Tushar Kumar Singh

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Education

Guru Gobind Singh Indraprastha University (GGSIPU), EDC, Aug 2025 - Jul 2029
B.Tech. in Artificial Intelligence and Machine Learning, CGPA 8.86/10
Coursework: Data Structures, Algorithms, Computer Networks, Operating System, Database Management System.

Technical Skills

Languages: Python, C, C++, Java, Bash, HTML, CSS, JavaScript, TypeScript
ML & AI: Deep Learning, Computer Vision, RAG, Agentic AI, LangChain, LangGraph
ML Libraries: PyTorch, TensorFlow, Scikit-learn, OpenCV, MediaPipe, NumPy, Pandas, Matplotlib
Web, Backend & Databases: FastAPI, React, Node.js, Express.js, REST APIs, PostgreSQL, MySQL, MongoDB
Cloud & DevOps: Git, GitHub, Linux, AWS, Docker, GitHub Actions (CI/CD), Kubernetes

Experience

Fiverr (Remote) - Freelance Software Developer (Unreal Engine, C++)
- Delivered 9 end-to-end projects for 7 international clients within tight deadlines, owning requirements, implementation in C++ and Blueprints, and handoff.
- Debugged and resolved crashes, logic errors, and performance bottlenecks across 3 production projects, improving stability across devices.

Projects

3D Brain Tumor Segmentation | Python, TensorFlow, 3D U-Net, Swin-UNETR, FastAPI (Jul 2026 - Aug 2026)
- Built an end-to-end 3D segmentation pipeline on BraTS 2019 multi-modal MRI, benchmarking a 3D U-Net baseline against a Swin-UNETR using 128x128x128 patch training and sliding-window inference.
- Designed a class-weighted Dice + Cross-Entropy loss; the 3D U-Net reached 0.85 Whole Tumor Dice (vs. 0.74 for Swin-UNETR), with per-class Dice of 0.77 / 0.84 / 0.80 on necrotic core, edema, and enhancing tumor.
- Deployed both models behind a FastAPI REST API (per-request model selection, weights from Hugging Face Hub) with a Streamlit UI for MRI upload, slice-wise overlays, and downloadable masks.

Neural Flow Dynamics | Python, FastAPI, PostGIS, Docker Compose, React, TypeScript (Nov 2026 - Present)
- Designed a research platform for weather forecasting and disaster decision support with a six-stage pipeline (acquire, validate, forecast, assess hazards, plan, approval) on versioned interfaces and full data provenance.
- Implemented a source-independent ingestion core with idempotent, provenance-tracked manifests, Alembic migrations, and a source-feasibility checker for ERA5 and NOAA GFS.
- Containerized the FastAPI + React/TypeScript + PostgreSQL + Redis stack with Docker Compose; enforced quality with pytest, Ruff, mypy, and ESLint. Designed a ClimODE-inspired neural-ODE forecaster.

Agentic-GameAI | Python, LLM APIs, Gymnasium, ALE, gym-sokoban, Pygame (Oct 2026 - Nov 2026)
- Built a modular framework for LLM agents that play five games of rising difficulty, reusing a shared core (retrying LLM client, JSONL logging, Pygame viewer).
- Implemented Sokoban planning with deadlock detection, retries from failed attempts, and an optional BFS hint; added switchable prompt hints to measure their effect.
- Built a hybrid real-time Breakout agent (pixel-to-state extraction, reflex layer, background LLM strategy) and a multi-agent Snake Arena against algorithmic opponents.

Achievements & Activities

- i7 Lead, IoSC GGSIPU-EDC: lead a team that builds AI/ML and game projects and mentors its members.
- Hackathons: Top 10 of 2,000+ teams at Fintechstico (NSUT), Top 50 of 2,700+ at Brainwave (GGSIPU), Top 50 of 1,500+ at AzinHack (GGSIPU).
- Solved 300+ LeetCode problems across data structures and algorithms; contest rating 1400+.